FIITJEE

ALL INDIA TEST SERIES

FULL TEST – V

JEE (Advanced)-2020

PAPER – 2

ANSWERS, HINTS & SOLUTIONS

Physics

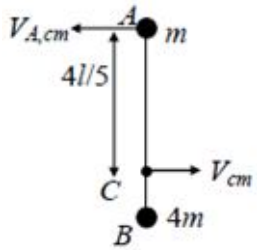
PART – I

SECTION – A

1. D

Sol. $J = 5m \times V_{cm}$

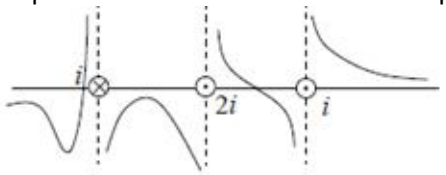
In the COM frame, particles will be in circular motion.



$$T = \frac{m\left(V_{A,cm}\right)^2}{(4l/5)} = \frac{m\left(\frac{J}{5m}\right)}{(4l/5)} = \frac{J^2}{20ml}$$

2. D

Sol. If upward direction of *B* is considered positive the variation can be shown as



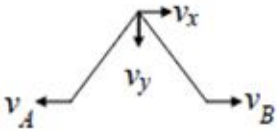
3. B

Sol. $v_x \cos 30^o + v_y \cos 60^o = v_B \cos 30^o$

$v_x\sqrt{3} + v_y = 12.....(1)$

$v_y \cos 30^o - v_x \cos 60^o = v_A \cos 60^o$

$v_y\sqrt{3} - v_x = 8.....(2)$



Solving (i) and (ii), we get

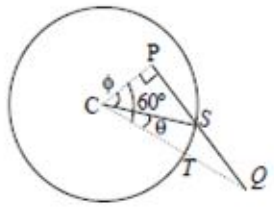
$v_x\left(3\sqrt{3}-2\right)m/s \; and \; v_y = \left(3+2\sqrt{3}\right)m/s$

$\therefore v_C = \sqrt{v_x^2 + v_y^2} = 2\sqrt{13}m/s$

4. D

Sol. $\tan 60^o = \dfrac{PQ}{CP}$

$\sqrt{3}CP = \dfrac{\sqrt{3}R}{\sqrt{2}}$



$CP = \dfrac{R}{\sqrt{2}}$

$CQ = \sqrt{\dfrac{3R^2}{2} + \dfrac{R^2}{2}} = \sqrt{2}R$

$\cos\phi = \dfrac{CP}{CS} = \dfrac{R}{\sqrt{2}R} = \dfrac{1}{\sqrt{2}}$

$\phi = 45^o \Rightarrow \theta = 15^o; PS = \dfrac{R}{\sqrt{2}}$

area of CPST = area ΔCPS + area of sector CST

$\left(\dfrac{1}{2}\times\dfrac{R}{\sqrt{2}}\times\dfrac{R}{\sqrt{2}}\right) + \dfrac{R^2}{2}\left(15^o \times \dfrac{\pi}{180^o}\right)$

$= \dfrac{R^2}{4} + \dfrac{R^2\pi}{24} = \dfrac{R^2}{24}(\pi+6)m^2$

$emf = A\dfrac{dB}{dt} = \dfrac{R^2}{24}(\pi+6)\times 4 = \dfrac{R^2}{6}(\pi+6)V$.

5. A

Sol. $p = mv = 0.02 \times 2000 \times 10 = 400N - s$

6. D

Sol. Let V be the volume of liquid flown out of container

$$V = \frac{AL}{3}\left[\gamma\Delta T + 2\gamma\Delta T + 3\gamma\Delta T\right] - \left[A\left(1+\gamma\Delta T\right)L\left(1+\frac{\gamma}{2}\Delta T\right) - AL\right]$$

$$= 2AL\gamma\Delta T - AL\left(1+\frac{3\gamma}{2}\Delta T\right) + AL = \frac{AL\gamma\Delta T}{2}$$

7. A

Sol. At position 1, $q = 0$, $t = 0$

$$q = CV_0\left(1-\cos\omega t\right)$$

At $t = \frac{\pi}{2\omega}, q = CV_0\left(1-\cos\omega\frac{\pi}{2\omega}\right) = CV_0 = q_0$

$$t' = t - \frac{\pi}{2\omega}, t' \geq 0$$

$$q = q_0\cos\omega t' = q_0\cos\left(\omega t - \frac{\pi}{2}\right) = q_0\sin\omega t$$

In position $2: q = q_0\sin\omega t \ for\, t \geq \frac{\pi}{2\omega}$

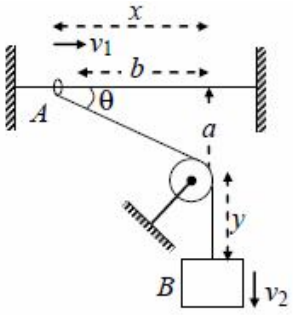
$$i = \omega q_0\cos\omega t = CV_0\omega\cos\omega t$$

8. AD

Sol. By constraint relation, $y+\sqrt{x^2+a}=$constant

$$\frac{dy}{dt} + \frac{2x}{2\sqrt{x^2+a^2}}\frac{dx}{dt} = 0$$

$$v_2 + \cos\theta\left(-v_1\right) = 0$$

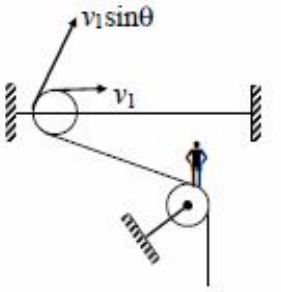


$$v_2 = v_1\cos\theta$$

$$\frac{dv_2}{dt} = \frac{dv_1}{dt}\cos\theta - v_1\sin\theta\frac{d\theta}{dt}$$

$$a_2 = a_1\cos\theta - v_1\sin\theta\frac{d\theta}{dt}$$

If ring is observed with respect to an observer at pulley,



$$\omega = \frac{v_1 \sin\theta}{\sqrt{a^2 + b^2}}$$

9. ABCD

Sol. If h<<H

Orbit velocity $= \sqrt{gr} = 8km/s$

Escape velocity $= \sqrt{2gr} = 11.2km/s$

If $v <$ *ve*, total energy is negative and trajectory will be bound.

If $v \geq$ *ve*, total energy is zero or positive and trajectory will be unbound.

10. ABCD

Sol. $K_{\max} = eV_s = hf - \phi$

$$\Rightarrow V_s = \frac{hf}{e} - \frac{\phi}{e}$$

$$K'_{\max} = eV'_s = nhf - \phi = n\left(eV_s + \phi\right) - \phi$$

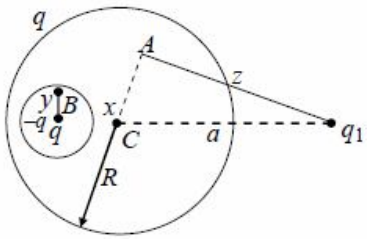
$$eV'_s = neV_s + \left(n-1\right)\phi \quad \left(n > 1\right)$$

$$V'_s = nV_s + \frac{\left(n-1\right)\phi}{e}$$

$$V'_s - V_s = \left(n-1\right)\left(V_s + \frac{\phi}{e}\right)$$

11. ABCD

Sol. The charge distribution is shown in figure. Due to q_1, the charge induction is non – uniform but net induced charge due to q_1 is zero. At point *A* field is zero and potential is equal to the potential at center.



$$V_A = V_C = \frac{1}{4\pi\varepsilon_0}\frac{q}{R} + \frac{1}{4\pi\varepsilon_0}\frac{q_1}{a}$$

At point *B*, field is due to charge *q* only

$$E_B = \frac{1}{4\pi\varepsilon_0}\frac{q}{y^2}$$

At point *B*, potential is due to charge *q* at center of cavity, *-q* induced on inner surface of cavity in addition to the potential at the center.

$$\therefore V_B = \frac{1}{4\pi\varepsilon_0}\left[\frac{q_1}{a}+\frac{q}{R}+\frac{q}{y}-\frac{q}{r}\right]$$

12. AC

Sol. Minima will be formed at *O*, if $SS_1 + S_1O - SS_2 - S_2O = \frac{n\lambda}{2}, n = 1, 2, ...$

For minimum value of d, n=1

$$\therefore \frac{\lambda}{4} = \sqrt{1+d^2} - 1$$

$$\left(1+d^2\right)^{1/2} - 1 = \frac{\lambda}{4}$$

$$1+\frac{d^2}{2} - 1 = \frac{\lambda}{4}(Negative\, smaller\, terms)$$

$$\Rightarrow d = \sqrt{\frac{\lambda}{2}};\ \beta = \frac{D\lambda}{d} = \sqrt{2\lambda}$$

13. CD

14. AB

Sol. At point Q, slope is zero always, so there will not be any potential energy stored at Q

15. C

Sol. $\vec{p}_i = 1\left(8i+6\hat{j}\right)+2\left(5\hat{i}-5\hat{j}\right) = 18\hat{i}-4\hat{j}$

$$\vec{p}_f = \vec{f}_{ext}.\Delta t + \vec{p}_i = -60\hat{j}+18\hat{i}-4\hat{j} = 18\hat{i}-64\hat{j}$$

16. D

Sol. Initial velocity of centre of mass

$$u_{cm} = \frac{\vec{p}_i}{m_1+m_2} = \frac{18\hat{i}-4\hat{j}}{3} = 6\hat{i}-\frac{4}{3}\hat{j}$$

$$\therefore \vec{r}_{cm(t=1)} = \vec{r}_{cm(t=0)} + \vec{u}_{cm}(1) + \frac{1}{2}a_{cm}(1)^2$$

$$= 20\hat{j}+\left(6\hat{i}-\frac{4}{3}\hat{j}\right)+\frac{1}{2}\left(-10\hat{j}\right) = 6\hat{i}+\frac{41}{3}\hat{j}$$

17. D

Sol. $k_1, k_2\ and\ k_3$ are closed

$$\text{Time constant} = \left(\frac{12\times 6}{12+6}+6\right)R\times 2C = RC$$

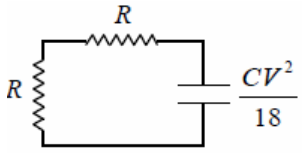
Initial current in $R_2 = \frac{V}{30R}$

Final current in $R_2 = \frac{V}{18R}$

Current in R_2 at an instant $t = \frac{V}{18R} + \left(\frac{V}{30R} - \frac{V}{18R}\right) e^{\frac{t}{20RC}}$

18. A

Sol. When k_1, k_3 opened, k_4 closed



Heat $= \frac{CV^2}{36}$.

Chemistry

PART – II

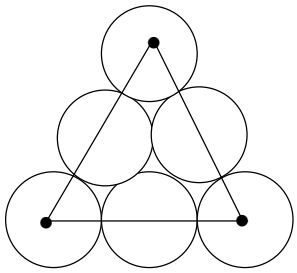
SECTION – A

19. B

Sol. Chromyl chloride test is shown by ionic chlorides. $K_2[PtCl_6]$ and $Cl-Hg-Hg-Cl$ have covalently bonded chlorine atoms.

20. D

Sol.



21. C

Sol. Area under curve = charge passed = 2 coulombs

$$w = Zit = Z = \frac{w}{q} = \frac{100}{2} = 50$$

For constant current; $q = it \Rightarrow 2 = i \times 30$

$$i = \frac{2}{30} A = \frac{200}{3} mA$$

22. C

23. D

Sol. Possible binary solutions are, Benzene + Toluene; Benzene + Xylene; Toluene + Xylene and ternary solution is Benzene + Toluene + Xylene.

In equimolar solutions, mole fractions are 1/2 in binary solution and 1/3rd in ternary solution for each component.

$\therefore$ For Benzene + Toluene, $P = \frac{1}{2}(75) + \frac{1}{2}(22) = \frac{97}{2} = 48.5$ *or* $48\frac{1}{2}$

For Benzene + Xylene $P = \frac{1}{2}(75) + \frac{1}{2}(10) = \frac{85}{2} = 42.5$

For Xylene + Toluene $P = \frac{1}{2}(22+10) = 16$

For Ternary solution $P = \frac{1}{3}(75+22+10) = \frac{107}{3} = 35\frac{2}{3}$

Hence, option (D) is not possible

24. D

Sol. $P_4 + 3NaOH + 3H_2O \rightarrow PH_3 + 3NaH_2PO_2$

Mole of $NaH_2PO_2 = 0.6$

25. D

Sol. More conjugated product is formed.

26. ABC

Sol. $Si + CH_3Cl \xrightarrow{Cu\ powder,\ 570K} (CH_3)_2SiCl_2 \xrightarrow{H_2O} (CH_3)_2Si(OH)_2 \xrightarrow{Polymerization} [(CH_3)SiO]_n$

$\downarrow O_2$

$SiO_2 \xrightarrow{Excess\ of\ coke,\ electric\ arc} SiC + CO$

27. AC

Sol. S=KOH

$P=I_2 \downarrow$

- $C_{17}H_{35}COOK$ *is a 'soft soap'*
- $KI(aq) + I_2(s) \longrightarrow KI_2(aq)$

28. A,C

Sol. Monoclinic : $\alpha=\gamma=90^0$, $\beta \neq 90^0$ so depends on β.

Rohmbohedral: $\alpha \neq \beta \neq \gamma$ so depends on angles.

29. BCD

Sol. $\Delta T_f = i.K_f.m$

$0.558 = i \times 1.86 \times .1$

$i = 3$

$\therefore \left[Co(NH_3)_5 Cl\right]Cl_2$

30. BC

Sol. $H_2SO_5, H_2S_2O_8$ due to peroxy linkage want to convert into normal oxide and acts as good oxidising agent.

31. ABC

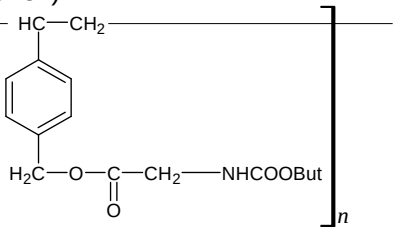
32. ABD

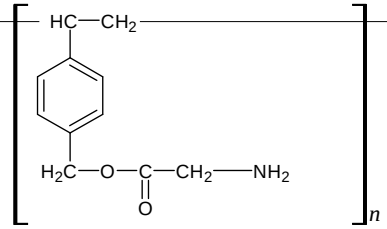
Sol. $E^0_{Cr^{3+}/Cr^{2+}} < E^0_{Fe^{3+}/Fe^{2+}} < E^0_{Co^{3+}/Co^{2+}}$

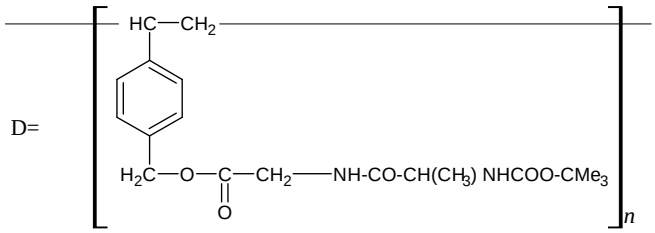
33. A

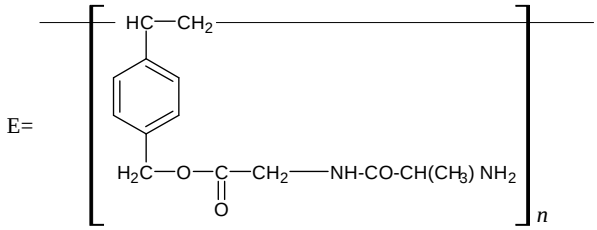
34. A

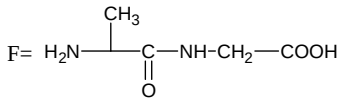
Sol. (for Q. 33 -34)

B= 

C= 

D= 

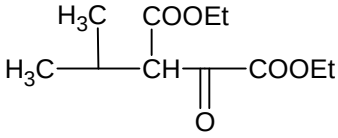
E= 

F= 

35. A

36. A

Sol. (for Q. 35 – 36):

A= 

Mathematics

PART – III

SECTION – A

37. B

Sol. $\frac{f''(x)}{f'(x)} - \frac{g''(x)}{g'(x)} = \frac{\frac{d}{dx}(f'(x)/g'(x))}{\frac{f'(x).g'(x)}{(g'(x))^2}} = \frac{g'}{f'}.\frac{d}{dx}\left(\frac{f'}{g'}\right) = \frac{2f'(x)}{f(x)}\left[\because f(x).g(x) = 1\right].$

38. B

Sol. Let $f^{-1}(x) = t \Rightarrow f(t) = x$

$\int_{f(a)}^{f(b)} 2x\left(b - f^{-1}(x)\right)dx = \int_a^b 2f(t).(b-t).f'(t)dt$ = b $[(f(t))^2]_a^b$ - b $\int_a^b$ t .2f(t) .f '(t) dt

$= \int_a^b \left[(f(x))^2 - (f(a))^2\right]dx$

39. B

Sol. $y = \log_{1/3}\left(x - \frac{1}{3}\right) + \log_3\sqrt{9x^2 - 6x + 1} = \log_{1/3}\left(x - \frac{1}{3}\right) + \log_3|3x - 1| = 1.$

Since $x > \frac{1}{3}$, maximum {AB} = 3.

40. C

Sol. $a_n = \frac{\frac{3n}{2}\left[a + (3n-1)d\right]}{\frac{2n}{2}\left[a + (2n-1)d\right] - \frac{n}{2}\left[a + (n-1)d\right]} = 3; \quad \therefore \lim_{n\to\infty}\sum_{r=1}^{2n}\frac{1}{n}.3 = 6.$

41. D

Sol. $e^{2x} + 2e^x < a(e^x - 1) \Rightarrow (e^x - 1)^2 + 4(e^x - 1) + 3 < a(e^x - 1)$

$\Rightarrow (e^x - 1) + \frac{3}{e^x - 1} + 4 < a \Rightarrow a > 4 + 2\sqrt{3}.$

42. B

Sol. $2x^3y\,dy + 3x^2y^2dx = \frac{2yxdx - x^2dy}{y^2}$

$\int d\left(x^3y^2\right) = \int d\left(\frac{x^2}{\hat{j}}\right)$

43. A

Sol. $\frac{1-|x|}{2-|x|} \geq 0 \Rightarrow \frac{|x|-1}{|x|-2} \geq 0 \quad \Rightarrow$ *use wavy curve of* $\frac{(x-1)(x+1)}{(x-2)(x+2)} \geq 0$

44. BCD

Sol. $I = \int_{-1}^{1} \left(e^{x^3} + e^{-x^3} \right) dx > 2$

Graph is concave upwards

$\Rightarrow I < f(0) + f(1)$

45. ACD

Sol. $f'(x) = e^{-\frac{1}{x^2}} . \frac{2}{x^3} + \frac{\pi}{2} \sqrt{1 + \sin \frac{\pi x}{2}}$

$\lim_{x \to 0^+} \frac{e^{-\frac{1}{x^2}}}{x^3} = 0$

And

Also $\lim_{x \to \infty} f(x) = \infty$

f''(x) does not exist for x=3, 7, 11….

46. AD

Sol. $\frac{2x^4 + 2y^4 + 4z^4 + 1}{4} \geq 2xyz$

And $x^4 y + xy^4 + \frac{4}{x^2 y^3} + \frac{1}{x^3 y^2} + 8 \geq 5 - 2$

$\Rightarrow \ell_2 > 10$ (equality cannot hold)

47. AB

Sol. (Put $x = y + \frac{1}{2}$)

48. AB

Sol. $\left(\frac{1}{x-1} + 1 \right) - \left(\frac{4}{x-4} + 1 \right) - \left(\frac{5}{x-5} + 1 \right) + \left(\frac{8}{x-8} + 1 \right) = \frac{6x^2 - 27x}{40}$

$\Rightarrow x = 0$

Also $\frac{2x-9}{(x-1)(x-8)} - \frac{2x-9}{(x-4)(x-5)} = \frac{3}{40}(2x-9)$

And x=9

49. ABD

Sol. Condition is satisfied for (a, b+c-2a)

$\Rightarrow b + c = 2003$

49. ABD

Sol. Condition is satisfied for (a, b+c-2a)

$\Rightarrow b + c = 2003$

50. BC

Sol. Since tan ($5\pi \cos \theta$) = cot ($5\pi \sin \theta$)

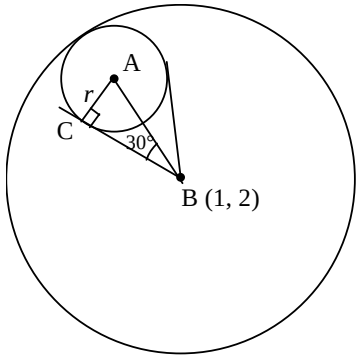
Or $\sin\theta + \cos\theta = \frac{2n+1}{10}$, $n \in I$

$\Rightarrow -1 < \frac{2n+1}{10\sqrt{2}} < 1 \Rightarrow -\left(\frac{10\sqrt{2}+1}{2}\right) < n < \left(\frac{10\sqrt{2}-1}{2}\right)$

hence total number of solution is 14.

51. B

Sol. Let r & R be radius of required & given circle resp. & let centre is (h, k).



By given condition

$\sqrt{(h-1)^2 + (k-2)^2} = R - r$

Now, $\frac{r}{AB} = \sin 30°$

$r = AB/2$

$= (R - r)/2 \qquad (AB = R - r) \Rightarrow r = \frac{R}{3}$

$\Rightarrow \sqrt{(h-1)^2 + (k-2)^2} = R - \frac{R}{3} = \frac{2R}{3}$

Now, $R = \frac{3\sqrt{3}}{2}$

$\Rightarrow \sqrt{(h-1)^2 + (k-2)^2} = \frac{2}{3} \times \frac{3\sqrt{3}}{2} = \sqrt{3}$

$\therefore$ Locus is $(x-1)^2 + (y-2)^2 = 3$ which is C.

52. D

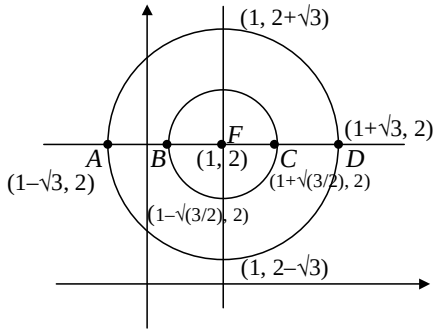
Sol. When major axis of ellipse (E) is parallel to x–axis then equation of E will be

$$\frac{(x-1)^2}{a^2} + \frac{(y-2)^2}{b^2} = 1 \quad (1)$$

Such that $a^2 + b^2 = 3$

In this case one focus will lie in between F and D and other in between A and F.

i.e., $0 < |x-1| < \sqrt{3}$ and $y = 2$.



53. C

Sol. If one number is drawn from the set {1, 2, 3…….n} the probability α that it is divisible by 2 is given

by $\alpha = \frac{\left[\frac{n}{2}\right]}{n}$.

$\Rightarrow$ P (the number is not divisible by 2) = 1 - α.

Now, $x^2 - y^2 = (x - y)(x + y)$

$\therefore P_2 = \alpha \cdot \alpha + (1 - \alpha)(1 - \alpha) = 1 - 2\alpha + 2\alpha^2$

54. D

Sol. Prob. β that the number is divisible by 3 = $\frac{[n/3]}{n}$

$\therefore$ P (the number is not divisible by 3) = 1 - β.

$\therefore P_3 = \beta^2 + (1 - \beta)^2 = 1 - 2\beta + \beta^2$